

Equal-time Limber κ_3 vertex: complete math flow (matter $P(k) \rightarrow \zeta$)

`scripts/sachs_sft/callables/kappa3_vertex/equal_time_limber`

Traced from the canoes source (2026-06-02) and cross-checked against the STF_lensing paper

This note traces the *entire production path* that turns the bare matter power spectrum $P_\delta(k, z=0)$ into the equal-time third-cumulant table ζ_{XYZ} that the sft-wick non-local K -vertex consumes. Every computational step is paired with (a) the equation it implements, (b) the exact `file:line` in code, and (c) the authoritative paper equation it corresponds to.

Symbols. χ comoving distance, λ affine projection parameter, ℓ multipole, γ angular separation, Φ_{00}/Ψ_0 the two driving fields, H_0 Hubble, Ω_m , D linear growth, A Poisson amplitude, $\mathcal{H}$ = conformal Hubble $\mathcal{H}$.

Roots. Code CAN = `/Users/zzhang/projects/canoes/src/canoes`; paper PAP = `.../STF_lensing/sections`.

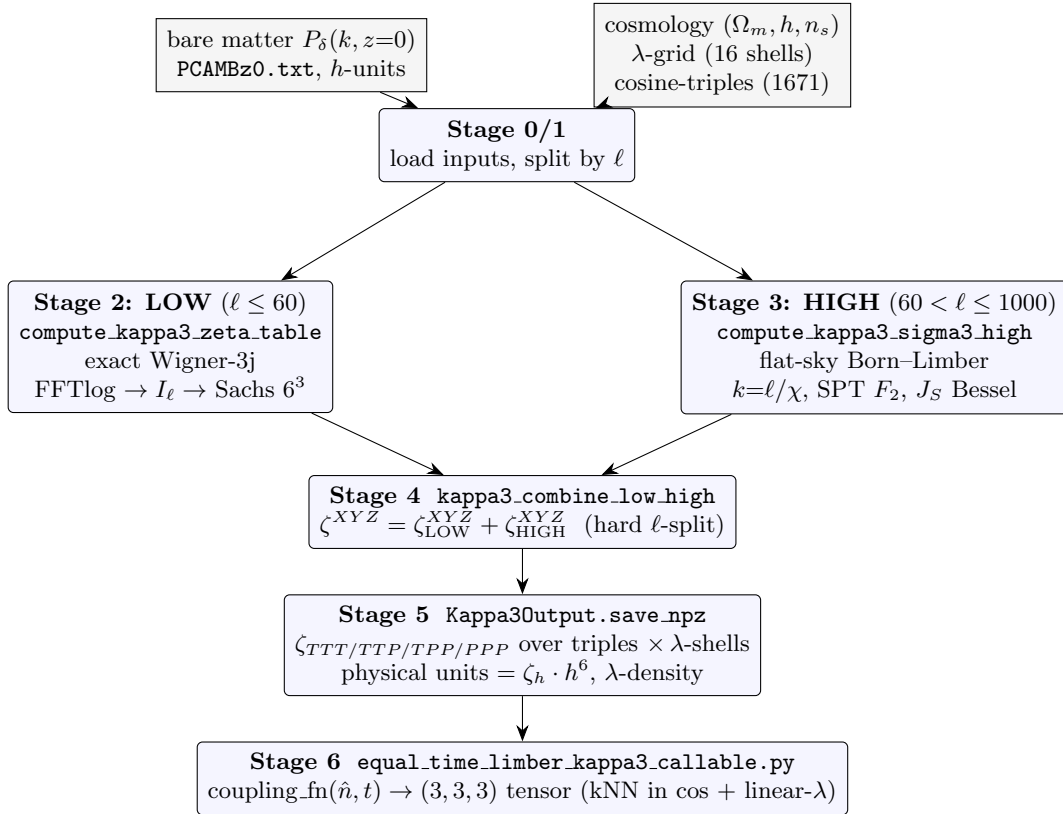
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1 Pipeline at a glance

The build script `build.equal_time_limber_table.py` (this folder) is only the conductor: it passes arguments and sums two branches (`build:133--151`).



2 Stage 0 — Inputs

2.1 Matter power spectrum (the single field-theory primitive)

$$\begin{aligned}
&P_\delta(k, z=0) \text{ from } \texttt{PCAMBz0.txt} \quad [k \text{ in } h/\text{Mpc}, P \text{ in } (\text{Mpc}/h)^3] \\
&\text{low-}k : \quad P(k) = P(k_{\min}) (k/k_{\min})^{n_s}, \quad n_s = 0.97 \\
&\text{high-}k : \text{ raises; } \quad \text{in range: log-log linear interpolation.}
\end{aligned} \tag{1}$$

code: `_local_cosmo_pk.py:42--53` (feeds the bare matter P_δ , not P_Φ); `class _CambTablePk` `CAN/cosmo/pk.py:85--154`.

paper: $P_\Phi(k; z=0)$ is named the single time-independent primitive (`PAP/appendix.tex:496--500`); link $P_\delta(k) = [k^2/A(a=1)]^2 P_\Phi(k)$ in eq: `appendix bphi tree (appendix.tex:608--614)`.

2.2 Cosmology

$$\Omega_m = \frac{\Omega_c h^2 + \Omega_b h^2}{h^2} = \frac{0.12029 + 0.02207}{0.6711^2} = 0.3160919980475834, \quad h = 0.6711, \quad n_s = 0.97. \tag{2}$$

code: `_local_cosmo_pk.py:33--39`; defaults `CAN/cosmo/background.py:42--51`.

2.3 Radial grid (λ -shells) and angular grid (cosine-triples)

λ -shells: 16 nodes 396.6...2326.6 Mpc ($z = 0.1...5.7$), read from the L2 NPZ. Triples $(\cos \gamma_{12}, \cos \gamma_{23}, \cos \gamma_{31})$ on a 15-node cos-grid ($\cos = k/7$), triangle-filtered to 1671 realizable triples.

code: λ -grid `build:61--69`; triangle filter `build:72--81`.

3 Stage 2 — LOW branch (exact Wigner-3j, $\ell \leq 60$)

Entry `compute_kappa3_zeta_table` `CAN/sachs/kappa3.py:3777`. The vertex factorizes into a *radial* stage (build $b_{\ell_1 \ell_2 \ell_3}(\lambda)$) and an *angular* stage (Wigner-3j $\rightarrow \zeta$).

3.1 FFTlog decomposition of P_δ ($N_{\max} = 128$ Mellin modes)

$$P_\delta(k) \approx \sum_n c_n k^{\nu_n}, \quad \nu_n = \text{bias} + \frac{2\pi i n}{N_{\max} \Delta \kappa}, \quad \text{bias} = -1.6, \quad N_{\max} = 128. \tag{3}$$

$N_{\max}$ is the number of FFTlog k -samples (not a Bessel order, not the bandlimit $\ell_{\max}$). *code:* `CAN/nue11/transfer/fftlog.py:43--131`; half-spectrum build `kappa3.py:458--466`.

3.2 Radial double-Bessel transform (the I_ℓ kernel)

$$I_\ell(\nu, t) = 4\pi \int_0^\infty dv v^{\nu-1} j_\ell(v) j_\ell(vt), \quad (\text{ASZ-2017 Mellin integral}) \tag{4}$$

$$P_\ell(r; \chi) = \sum_n c_n \chi^{-\nu_n} \frac{I_\ell(\nu_n + \text{shift}, r/\chi)}{4\pi} \quad (\text{power-law "PL" leg}), \tag{5}$$

plus a closure leg $S_\ell^{\text{spec}}(r; \chi)$ (δ^2 SPT leg, Γ -regularized).

code: I_ℓ `kernels/il.py:1--6,1288`; PL leg `_bl_jax.py:679--698`; spec leg `_bl_jax.py:700--717`. *paper:* per-leg $\Delta_\ell^X(k; \lambda)$ is eq: `appendix delta ell (appendix.tex:562--568)`; per-leg kernel M'_ℓ is eq: `appendix Mprime (appendix.tex:653--659)`.

3.3 Tree-level SPT layout + the Sachs $6 \times 6 \times 6$ basis contraction

$$b_{\ell_1 \ell_2 \ell_3}(\lambda) = \left(\frac{2}{\pi}\right)^3 \sum_{k,l,m \in \text{basis}(6)} (C_a \cdot M)_k (C_b \cdot M)_l (C_c \cdot M)_m \times \sum_{\substack{3 \text{ cyc} \\ \text{spec}}} \sum_{6 F_2 \text{ mono}} \text{coeff} \int_0^\infty dr r^2 S_{\ell_c}^{\text{spec}}(r; \chi) P_{\ell_a}(r; \chi) P_{\ell_b}(r; \chi) \times \Delta_{\text{tri}}(\ell_1 \ell_2 \ell_3). \quad (6)$$

The $18 = (3 \text{ cyclic spectator}) \times (6 F_2 \text{ monomials})$ terms encode the SPT tree matter bispectrum $B_\delta = 2F_2 PP + \text{cyc}$; (C_a, C_b, C_c) are the per-leg Sachs `CoefVecs` (§3.6), M the lightcone derivative basis (§3.5).

code: r -integral leaf `_bl_jax.py:914--917`; 6^3 contraction `_bl_jax.py:1081--1087`; SPT monomials `_phi_accumulator.py:49--69`. *paper:* this is exactly eq: `appendix b prime (appendix.tex:667--675)`, $B_{LL\ell_3}^{XYZ} = (2/\pi)^3 \int dr r^2 \sum_{t=0}^{17} c_t M_\ell^{\text{spec}} M_\ell^{\text{PL}} M_\ell^{\text{PL}}$.

3.4 The cosmology prefactor stack (applied per λ -shell)

The dimensionless [angular $\times$ SPT-shape] integral is multiplied by, in order:

#	factor	equation	code
i	Poisson ³	$A(a)^3 = \left[-\frac{3}{2} H_0^2 \Omega_m (1+z)\right]^3$, $H_0 = 100/c$ (h -indep.)	<code>kappa3.py:85,4222--4224</code> ; <code>cubed :4268--4269</code>
ii	Sachs geometric	$((1+z)^P)^3 = (1+z)^{12}$, $P = \text{SACHS_GEOMETRIC_POWER} = 4$	<code>kappa3.py:499</code> ; <code>sachs_driver.py:97</code>
iii	per-leg M	$s_X = C_X \cdot M$, $M[0]=D(1+z)$, $M[1]=-\partial_\chi[D(1+z)]$, $M[2]=\partial_\chi^2[\dots]$	<code>kappa3.py:482--512</code> ; <code>lightcone.py:342--350</code>
iv	spectator growth	$\times D(z) = M[0]/(1+z)$ (promotes $D^3 \rightarrow D^4$)	<code>kappa3.py:4495--4506</code>
v	radial measure	λ -density $\Rightarrow \times 1$ (<i>no</i> $(d\chi/d\lambda)$ Jacobian)	<code>kappa3.py:1761--1805,4508--4520</code>
vi	units	physical $\Rightarrow \zeta = h^6 \zeta_h$, $\chi \rightarrow \chi/h$, $\lambda \rightarrow \lambda/h$	<code>kappa3.py:4522--4530</code>

paper: per-leg $A(a) = -\frac{3}{2} H_0^2 \Omega_m / a$ and per-leg $(1+z)^4$ Sachs weight, `appendix.tex:603--604,536--543`; measure $d\lambda = a^2(\chi) d\chi$ in eq: `cumulant 2 measure change (cosmology.tex:979--988)`; $\bar{D} = a\chi$ in eq: `D equals a chi (appendix.tex:286--289)`.

Warning. The redshift scaling is *not* a clean power (see §8). The growth is clean (D^4), but the $(1+z)$ content is heterogeneous because `COEFS_PHI00` contracts $M[0], M[1], M[2]$ together with the $\mathcal{H}^2/\mathcal{H}'/L^2\chi^{-2}$ operator terms.

3.5 The lightcone derivative basis $M[0..2]$

$$M[0] = D(z)(1+z) = D/a, \quad M[1] = -\frac{d}{d\chi}[D(1+z)], \quad M[2] = \frac{d^2}{d\chi^2}[D(1+z)], \quad (7)$$

with $\mathcal{H} = E(z)/[(1+z)(c/100)]$, $E(z) = \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$. *code:* `CAN/cosmo/lightcone.py:328--329,342--350`.

3.6 The per-leg Sachs CoefVecs (what makes T vs P)

$$\text{COEFS_PHI00} = \left(\frac{L^2}{\chi^2} + 2\mathcal{H}^2 - 2\mathcal{H}', -\frac{2}{\chi} + 2\mathcal{H}, -1, 0, -1, 2 \right) \quad (T = \Phi_{00}, \text{ spin } 0), \quad (8)$$

$$\text{COEFS_PSI0} = \left(-\frac{\sqrt{L^2(L^2-2)}}{\chi^2}, 0, 0, 0, 0, 0 \right) \quad (P = \Psi_0, \text{ spin } 2), \quad (9)$$

with $L^2 = \ell(\ell+1)$. A leg's operator value is $g_{\text{leg}}(\ell, \chi) = \sum_{k=0}^5 \text{COEFS}[k] M[a_k]$, the gather $a_k = [0, 0, 0, 1, 2, 1]$. *code:* `CAN/nuell/transfer/sachs_driver.py:74--90`; gather `kappa3.py:482--484`; $L^2 = \ell(\ell+1)$ `basis.py:45--52, 111--120`.

3.7 Angular stage — exact Wigner-3j $\rightarrow \zeta$

For each cosine-triple with channel spins (s_1, s_2, s_3) :

$$\begin{aligned} \zeta_{\text{LOW}}^{XYZ}(\gamma_{12}, \gamma_{23}, \gamma_{31}; \lambda) = D(z) \sum_{\ell_1 \ell_2 \ell_3 \leq 60} b_{\ell_1 \ell_2 \ell_3}(\lambda) h \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ 0 & 0 & 0 \end{pmatrix} \sqrt{\frac{2\ell_1+1}{4\pi}} (-1)^{s_1} \\ \times \sum_{m_2} \begin{pmatrix} \ell_1 & \ell_2 & \ell_3 \\ -s_1 & m_2 & s_1 - m_2 \end{pmatrix} {}_{s_2} Y_{\ell_2 m_2}(\gamma_{12}, 0) {}_{s_3} Y_{\ell_3, s_1 - m_2}(\gamma_{31}, \varphi_3), \end{aligned} \quad (10)$$

with $h = \sqrt{(2\ell_1+1)(2\ell_2+1)(2\ell_3+1)/4\pi}$. Two Wigner-3j symbols appear: the parity/triangle anchor $(\ell_1 \ell_2 \ell_3; 0, 0, 0)$ (kept for *all* four channels) and the m -resolved $(\ell_1 \ell_2 \ell_3; -s_1, m_2, s_1 - m_2)$. Angular separations enter through ${}_s Y$ with $\theta = \arccos \cos \gamma$; the leading $D(z)$ is the spectator growth.

code: `scalar (TTT) three_pt.py:516--645`; spin-weighted `_spin_aware_three_pt.py:15--289`; `GEMV _jax_kernels.py:1012--1032`. *paper:* this is eq: `appendix zeta squeezed (appendix.tex:684--704)`,

$$\begin{aligned} \zeta_{XYZ}(\gamma) = \sum_{\substack{L, \ell_3 \\ 2L+\ell_3 \text{ even}, \ell_3 \leq 2L}} \mathcal{G}(L, L, \ell_3) P_{\ell_3}(\cos \gamma) B_{LL\ell_3}'^{XYZ}, \\ \mathcal{G}(L, L, \ell_3) = \frac{(2L+1)^2(2\ell_3+1)}{(4\pi)^2} [w_{3j}(L, L, \ell_3; 0, 0, 0)]^2. \end{aligned} \quad (11)$$

The paper writes the squeezed (L, L, ℓ_3) specialization (Legendre $P_{\ell_3}(\cos \gamma)$, two rays coincident); the code's general m -sum reduces to it for that configuration.

4 Stage 3 — HIGH branch (flat-sky Born–Limber, $60 < \ell \leq 1000$)

Entry `compute_kappa3_sigma3_high CAN/sachs/kappa3.py:3406--3684`.

4.1 Flat-sky Limber collapse ($k = \ell/\chi$, single shell)

$$\begin{aligned} u = |\ell_2|, \quad v = |\ell_3|, \quad \varphi = \angle(\ell_2, \ell_3), \quad w = |\ell_1| = \sqrt{u^2 + v^2 + 2uv \cos \varphi}, \\ k_i = \frac{\ell_i}{\chi(\lambda)} \quad (\text{equal-}\chi \text{ Limber}), \quad \text{radial kernel} = \frac{1}{\chi^4}. \end{aligned} \quad (12)$$

(equal- χ Limber, no factor of a ; the $1/\chi^4$ is three $1/\chi^2$ collapsed by two equal- χ δ 's.) *code:* `quadrature kappa3.py:1608--1649` ($n_\ell=96, n_\varphi=64$); $k = \ell/\chi$:`3570--3572`; $1/\chi^4$:`3585`.

4.2 Tree-level SPT matter bispectrum (F_2 kernel)

$$B_\delta(k_1, k_2, k_3) = 2[F_2(k_1, k_2)P_1P_2 + F_2(k_1, k_3)P_1P_3 + F_2(k_2, k_3)P_2P_3], \quad (13)$$

$$F_2(k_a, k_b) = \frac{5}{7} + \frac{1}{2}\mu\left(\frac{k_a}{k_b} + \frac{k_b}{k_a}\right) + \frac{2}{7}\mu^2, \quad \mu = \cos(k_a, k_b), \quad P_i = P_\delta(k_i, z=0). \quad (14)$$

code: `F2 kappa3.py:1690--1702`; *B* assembly :3573--3579; internal cosines from the closed ℓ -triangle :3530--3540. *paper:* same SPT tree in eq: `appendix bphi tree (appendix.tex:608--614)`.

4.3 Cosmology factors (HIGH mirror of §3.4)

$$\times D(z)^4 \quad (B_\delta \propto PP \propto D^4), \quad R_{\text{channel}} = \prod_{\text{legs}} (\pm) A(a)(1+z)^4, \quad \text{prefactor } \frac{1}{(2\pi)^4}, \quad (15)$$

$A(a) = -\frac{3}{2}\Omega_m H_0^2(1+z)$; sign + for $\Phi_{00}(T)$, - for $\Psi_0(P)$. *code:* `growth4 :3580--3583`; $A(a)$:3498--3500; response product :1669--1687, applied :3589--3594; $1/(2\pi)^4$:3541.

4.4 Spin via the analytic orientation (Bessel) integral

$$\theta_{ij} = \arccos \cos \gamma_{ij}, \quad Q = u r_2 + v e^{-i\varphi} r_3, \quad S = s_1 + s_2 + s_3, \quad (16)$$

$$\alpha_{\text{channel}} = \text{Re} \left[2\pi i^S e^{iS \arg Q} J_S(|Q|) e^{i(s_1 \arg \ell_1 + s_3 \varphi)} \right] \quad (S=0 \Rightarrow \alpha = 2\pi J_0(|Q|)), \quad (17)$$

with $r_2 = \theta_{12}$, $r_3 = x_3 + iy_3$ (law of cosines). Spins: TTT (0, 0, 0), TTP (0, 0, 2), TPP (0, 2, 2), PPP (2, 2, 2); all four nonzero in HIGH. *code:* vertices `kappa3.py:1705--1717`; α -kernel :1720--1758; spins :1664--1666.

4.5 HIGH assembled

$$\zeta_{\text{HIGH}}^{XYZ}(\text{triple}, \lambda) = \frac{1}{(2\pi)^4} \frac{D^4}{\chi^4} R_{\text{channel}} \int du dv d\varphi (uv w_\ell w_\ell w_\varphi) B_\delta \alpha_{\text{channel}}, \quad (18)$$

integration region: $\ell_{\text{cut}} < \max(\ell_1, \ell_2, \ell_3) \leq \ell_{\text{hi}}$.

code: radial base :3585; weighted $\times\alpha$ sum :3595--3596; HIGH window :3515--3518.

5 Stage 4 — Combine LOW + HIGH

$$\zeta^{XYZ}(\text{triple}, \lambda) = \zeta_{\text{LOW}}^{XYZ}(\text{triple}, \lambda) + \zeta_{\text{HIGH}}^{XYZ}(\text{triple}, \lambda). \quad (19)$$

A plain elementwise array sum on the *identical* (cosine-triple, λ) grid. Hard split at ℓ_{cut} , no tapering: LOW does exact 3j for $\ell \leq \ell_{\text{cut}}$, HIGH does flat-sky for $\max \ell \in (\ell_{\text{cut}}, \ell_{\text{hi}}]$; the boundary $\max \ell = \ell_{\text{cut}}$ is in LOW only. Grid/units/measure compatibility is asserted before the add; `cosmo_meta` is merged. *code:* `kappa3.combine_low_high kappa3.py:3710--3774` (sum :3766--3771, grid checks :3741--3742).

6 Stage 5 — Output table

`Kappa3Output` holds `cosine_triples(M, 3)`, `lambda_shells_Mpc`, `chi_shells_Mpc`, `z_shells`, `zeta.TTT/TTP/TPP/PPP(M, N $\lambda$ )`, `ell_max`, `cosmo_meta`. `save_npz` writes them with *physical* units (λ, χ in Mpc; $\zeta = h^6 \zeta_h$) and λ -density measure (no Jacobian). *code:* `dataclass kappa3.py:164--237`; `save_npz` :187--206. The build then overwrites the metadata block (`build:152--170`).

7 Stage 6 — Callable consumption (what sft-wick sees)

$$\cos \gamma_{ij} = \hat{n}_i \cdot \hat{n}_j \xrightarrow{\text{sort desc}} \text{cKDTree query } (k=4 \text{ inverse-distance}), \quad \lambda_{\text{mean}} = \bar{t},$$

$$\zeta_{\text{ch}} = \frac{\sum_k w_k \zeta_{\text{ch}}[\text{node}_k, \lambda]}{\sum_k w_k}. \quad (20)$$

The $(3, 3, 3)$ tensor places ζ_{TTT} at $[0, 0, 0]$, ζ_{TTP} at the $\{0, 0, 1\}$ perms, ζ_{TPP} at $\{0, 1, 1\}$, ζ_{PPP} at $[1, 1, 1]$; all index-2 (B-mode) entries are zero. Because `already_R_contracted=False`, sft-wick applies the response R itself (this table is the bare vertex K). *code: callable.py:129--228*. See the `slices/` figures: a contact spike at the collapsed apex $\cos = (1, 1, 1) \sim 100\times$ above a flat wide-separation plateau, growing toward high z .

8 Redshift scaling of ζ (the honest statement)

$\zeta_{TTT}(\text{apex}, z)$ does **not** factor as a clean $(1+z)^P D^Q \times [z\text{-independent shape}]$.

- **Growth is clean:** $Q = 4$. Three per-leg M -factors (each linear in D) $\times$ one spectator D . No extra D hides (the χ -derivatives in $M[1], M[2]$ differentiate the *shape*, not the amplitude). Metadata `growth_scaling="D^4"` is correct.
- $(1+z)$ **is not clean**. Explicit factors give $(1+z)^{15} = (1+z)^{12} [\S 3.4 \text{ ii}] \times (1+z)^3 [\text{Poisson cube, } \S 3.4 \text{ i}]$. But $\sum_k \text{COEFS_PHI00}[k] M[a_k]$ mixes $M[0] = D(1+z)$ with $M[1], M[2]$ (χ -derivatives carrying $\mathcal{H}$ -content) and the $2\mathcal{H}^2 - 2\mathcal{H}'$, L^2/χ^2 terms in one slot. $\mathcal{H} = E(z)/[(1+z)(c/100)]$ is not a power of $(1+z)$, so the exact exponent is ill-defined.
- **Leading-term scaling** (keep only the $\text{ID} \cdot M[0]$ L^2/χ^2 piece, dominant at $\ell \gtrsim$ a few):

$$\zeta_{TTT}(\text{apex}, z) \approx \left[-\frac{3}{2} H_0^2 \Omega_m \right]^3 (1+z)^{15} D(z)^4 \chi^{-6} \prod_i \ell_i(\ell_i+1) \cdot [\text{Wigner-3j}] \cdot \left[\text{SPT} \int r^2 dr \text{ shape} \right]. \quad (21)$$

This explains the slices: ζ grows steeply toward high z (the $(1+z)^{15} D^4$ front factor) and is sharply peaked at the collapsed apex.

9 Review — production step $\leftrightarrow$ equation completeness

Every step in the production path, with its equation and a verdict on whether an equation is attached.

step	action	code (file:line)	paper	ok
S0.1	load matter P_δ	<code>_local_cosmo.pk.py:42--53;</code> <code>pk.py:85--154</code>	<code>appendix.tex:496–</code> <code>500,608–614</code>	✓
S0.2	cosmology / grids	<code>build:61--81</code>	—	✓
S2.1	FFTlog of P_δ	<code>fftlog.py:43--131</code>	(numeric)	✓
S2.2	double-Bessel I_ℓ	<code>il.py:1--6;</code> <code>_bl_jax.py:679--717</code>	<code>delta ell, Mprime</code>	✓

step	action	code	paper	ok
S2.3	SPT 18-term + 6^3 contraction	<code>_bl_jax.py:914--1087;</code> <code>_phi_accumulator.py:49--69</code>	b prime	✓
S2.4i	Poisson $A(a)^3$	<code>kappa3.py:4222--4224,4269</code>	cosmology.tex:405--407; appendix:603--604	✓ ^{R1}
S2.4ii	$(1+z)^{12}$ Sachs	<code>kappa3.py:499;</code> <code>sachs_driver.py:97</code>	appendix:536--543	✓
S2.4iii	per-leg M	<code>kappa3.py:482--512;</code> <code>lightcone.py:342--350</code>	Mprime	✓
S2.4iv	spectator D ($\rightarrow D^4$)	<code>kappa3.py:4495--4506</code>	(code conv.)	✓ ^{R3}
S2.4v	λ -density ($\times 1$)	<code>kappa3.py:1761--1805</code>	measure; $D=a\chi$	✓
S2.4vi	h^6 units	<code>kappa3.py:4522--4530</code>	(h -bookkeeping)	✓
S2.7	exact Wigner-3j angular	<code>three_pt.py:516--645;</code> <code>_spin_aware_three_pt.py:15--289</code>	zeta squeezed + $\mathcal{G}$	✓ ^{R2}
S3.1	flat-sky Limber $k=\ell/\chi, 1/\chi^4$	<code>kappa3.py:3570--3585,</code> <code>1608--1649</code>	(Limber)	✓
S3.2	SPT F_2 matter bispectrum	<code>kappa3.py:1690--1702,</code> <code>3573--3579</code>	bphi tree	✓
S3.3	HIGH cosmology factors	<code>kappa3.py:3498--3500,</code> <code>3580--3594, 1669--1687</code>	appendix:603--604	✓
S3.4	spin Bessel orientation J_S	<code>kappa3.py:1705--1758</code>	(flat-sky 3PCF)	✓
S4	LOW+HIGH sum	<code>kappa3.py:3710--3774</code>	(numerical ℓ -split)	✓
S5	NPZ save (units, measure)	<code>kappa3.py:164--206</code>	—	✓
S6	callable $\rightarrow (3, 3, 3)$	<code>callable.py:129--228</code>	sft-wick contract	✓

Verdict: every production step has a corresponding equation. Three points are code $\leftrightarrow$ paper *conventions to keep on the radar* (not gaps):

R1 — Poisson sign. cosmology.tex:405--407 writes $\Phi_{00} = +A\delta$ with $A > 0$; the appendix and the code use $A(a) = -\frac{3}{2}H_0^2\Omega_m/a$ ($A < 0$). Because A enters cubed (odd power), the sign reaches ζ . Code is consistent with the appendix; the main-text caption sign differs. A one-line paper reconciliation, not a code change.

R2 — spin-2 Wigner-3j. Both code and paper keep the *scalar* parity anchor $(\ell_1\ell_2\ell_3; 0, 0, 0)$ for all four channels including the spin-2 ($\Psi_0 = P$) legs (only the m -sum 3j carries the $-s_1/s_1-m_2$ shifts). Code and paper agree; flagged because both are the documented pragmatic convention, not the full $(\pm 2, \dots)$ triple-spin anchor.

R3 — explicit D^4 growth. The code multiplies an explicit linear growth $D(z)^4$ (3 per-leg + spectator). The `\input`-ed paper sections do not write a D^4 law; they encode time-dependence via the per-leg $A(a)(1+z)^4$ windows on the $z=0$ primitive P_Φ . Physically D^4 is the tree-level scaling ($B_\delta \propto PP \propto D^4$), so the two are consistent, but the explicit D^4 is a code-side bookkeeping choice (the explicit statement exists only in the untracked cosmology2.tex:966).

10 End-to-end composite (LOW, one channel)

$$\begin{aligned}
P_\delta(k, z=0) &\xrightarrow{\text{FFTlog}(N_{\text{max}}=128)} P_\delta(k) \approx \sum_n c_n k^{\nu_n} \\
&\rightarrow P_\ell(r; \chi) = \sum_n c_n \chi^{-\nu_n} I_\ell(\nu_n + \text{shift}, r/\chi)/(4\pi), \quad I_\ell(\nu, t) = 4\pi \int dv v^{\nu-1} j_\ell(v) j_\ell(vt) \\
&\rightarrow b_{\ell_1 \ell_2 \ell_3}(\lambda) = \left(\frac{2}{\pi}\right)^3 A(a)^3 (1+z)^{12} \sum_{\text{basis}} (C_a M)(C_b M)(C_c M) \\
&\quad \times \sum_{\text{spec}} \sum_{F_2} \text{coeff} \int dr r^2 S_{\ell_c}^{\text{spec}} P_{\ell_a} P_{\ell_b} \\
&\rightarrow \zeta_{\text{LOW}}^{XYZ} = D(z) \sum_{\ell_1 \ell_2 \ell_3 \leq 60} b_{\ell_1 \ell_2 \ell_3} h(\dots; 0, 0, 0) \sqrt{\frac{2\ell_1+1}{4\pi}} (-1)^{s_1} \\
&\quad \times \sum_{m_2} (\dots; -s_1, m_2, s_1 - m_2)_{s_2} Y_{s_3} Y \\
&\rightarrow (\times 1 \text{ } \lambda\text{-measure}; \times h^6 \text{ physical}); \quad \zeta^{XYZ} = \zeta_{\text{LOW}}^{XYZ} + \zeta_{\text{HIGH}}^{XYZ}.
\end{aligned}$$

Channels: $T = \Phi_{00}$ (spin 0, COEFS_PHI00); $P = \Psi_0$ (spin 2, COEFS_PSI0). TTT/TTP/TPP/PPP = (PHI00, PHI00, PHI00)/(PHI00, PHI00, PSI0)/(PHI00, PSI0, PSI0)/(PSI0, PSI0, PSI0), spins (0, 0, 0)/(0, 0, 2)/(0, 2, 2)/(2, 2, 2).

Code traced at CAN=/Users/zzhang/projects/canoes/src/canoes (2026-06-02). Authoritative paper equations: eq: appendix b prime, eq: appendix zeta squeezed (appendix.tex:667--704); Poisson cosmology.tex:405--407, 561--570; Sachs $(1+z)^4$ appendix.tex:536--543; measure cosmology.tex:979--988; $D=a\chi$ appendix.tex:286--289. Reproduce: python build.equal.time.limber.table.py.